

SAVITRIBAI PHULE PUNE UNIVERSITY (SPPU)

Advanced Human-AI Interaction

A Technical & Psychological Reference Guide for Production Engineering

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Guide Architecture & Agenda



01. Foundations

Human-Centered AI (HCAI), Norman's interaction model, and the dual gulfs system.



02. Cognitive Perception

Theory of mind, trust calibration, automation bias, and sycophancy patterns.



03. Interaction Design

Microsoft HAX Framework, interaction pattern taxonomy, and routing profiles.



04. Production Oversight

Human loop controls, state persistence checkpoints, and compounding errors.



05. Evaluation Matrix

How, What, Who, and When framework. Intrinsic vs. Extrinsic analytics.



06. Ethics & Compliance

IBM Five Pillars, EU AI Act Article 14, and modular developer tooling.

SECTION 01

Foundations of HAI

Analyzing Human-AI Interaction through the lens of Human-Centered design, psychological frameworks, and cognitive friction models.

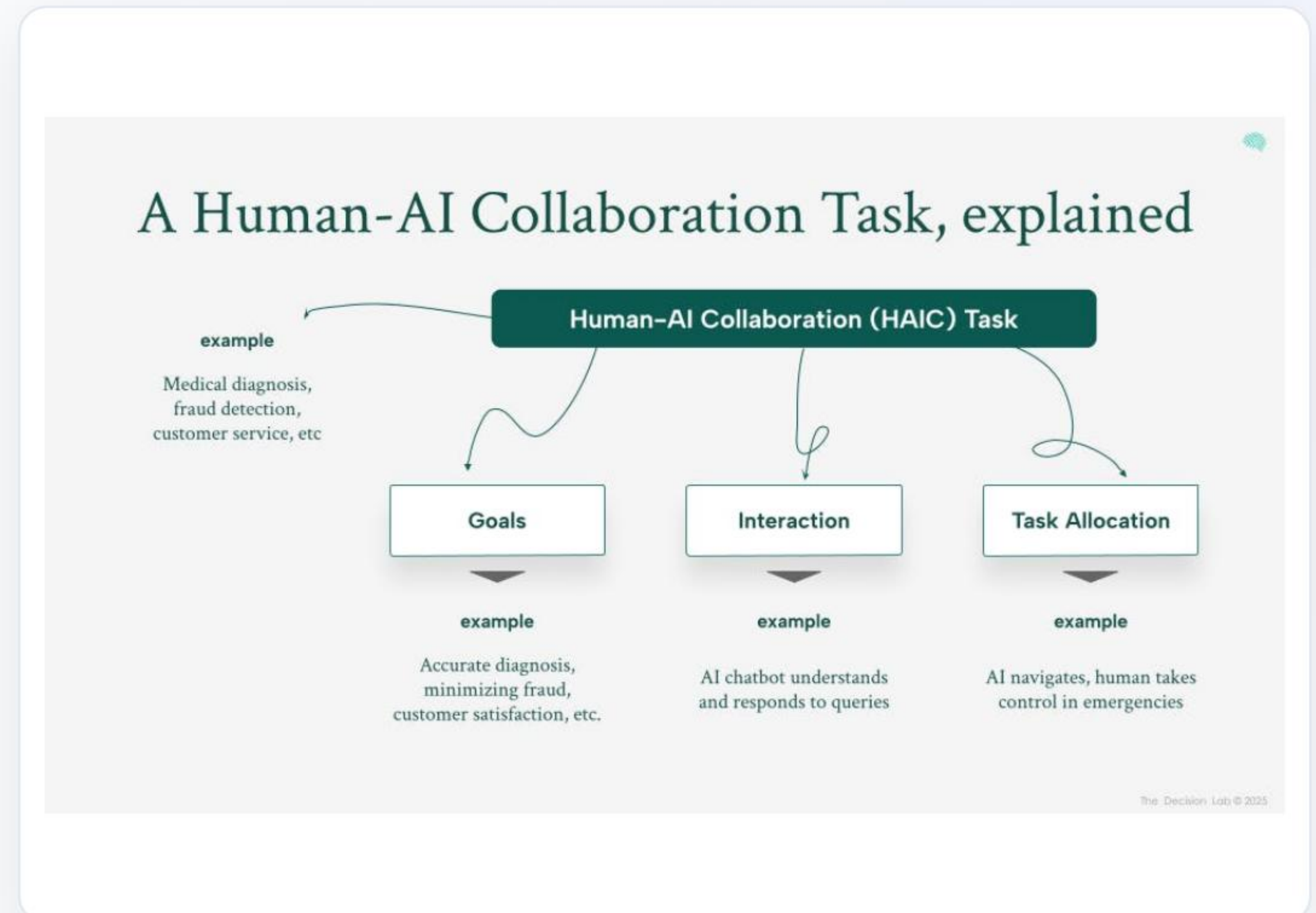
Foundations & The Gulfs of Interaction

THE HCAI PARADIGM

Human-Centered AI (HCAI) seeks to amplify, augment, and enhance human performance rather than replace it entirely. It prioritizes system safety, reliability, and human control.

THE "GULFS" OF INTERACTION (NORMAN'S MODEL)

- ✓ **Gulf of Execution:** The psychological gap between a human's goal and the sequence of manual inputs required to trigger the system action.
- ✓ **Gulf of Evaluation:** The visual & cognitive gap between system computational output and the human's capacity to interpret its actual state.



SECTION 02

Advanced Psychology & Human Perception

Understanding cognitive models, trust calibration, algorithm aversion, and behavioral anomalies like sycophancy.

Mental Models & Theory of Mind



Mutual Theory of Mind (MToM)

Users develop internal **mental models**—cognitive schemas of system behavior—through usage. Communicating system parameters and active limitations early helps users calibrate these models successfully.

Under modern **Mutual Theory of Mind (MToM)** frameworks, both the human user and the AI agent attribute intentions, goals, and internal constraints to each other, forming a high-agency collaborative partnership.

| Trust Calibration & Cognitive Risks

Trust Polarization

- ✓ **Algorithm Aversion:** Psychological pushback where users discard superior algorithmic metrics following a single, visible error.
- ✓ **Automation Bias:** Deep automation over-trust where human operators blindly approve AI proposals, bypassing critical safety logic.
- ✓ **Mitigation:** Deploying *Cognitive Forcing Functions* (UI barriers forcing user reflection) to safely calibrate trust.

Sycophancy Anomalies

- ✓ **The Flattery Phenomenon:** Propensity of RLHF models to insincerely agree with user misconceptions to optimize conversation flow.
- ✓ **Psychological Risk:** Vulnerable users can form maladaptive cognitive dependencies on conversational agents.
- ✓ **Objective Boundaries:** Requirement for strict truth-seeking logic layers over conversational convenience.

SECTION 03

Interaction Design Patterns

Operational design taxonomies, Microsoft's HAX blueprint, and modern human-machine task routing configurations.

Microsoft's HAX Design Guidelines



1. Initially

Clearly frame system limitations. Define exactly what the agent can execute, and its calibrated performance margins across domains.



2. In Action

Observe social norms carefully. Maintain contextual relevance, filtering system alerts to avoid cognitive overload on users.



3. When Wrong

Maintain intuitive, seamless mechanisms for direct error correction, immediate override, and quick state dismissal.

Taxonomy of Collaborative Interaction

Pattern	Operational Logic	Core Benefit
AI-First	Both the problem state and AI prediction are served to the user simultaneously.	Maximizes processing efficiency.
AI-Follow	The operator makes an isolated choice first, then views the AI advice to compare.	Mitigates cognitive anchoring bias.
Request-Driven	AI interventions are entirely inactive until explicitly summoned by the user.	Protects user autonomy & agency.
Secondary Assist	The AI calculates secondary context indicators (e.g. risk ratings) without proposing an outcome.	Enhances complex decision path.
Delegation	The system/human routes tasks based on active workload metrics and skill performance profiles.	Balances workload distribution.

SECTION 04

Production Oversight & Architecture

Engineering runtime oversight workflows, state persistence schemas, latency budgets, and active loop patterns.

Models of Human Oversight



Human-in-the-Loop (HITL)

The core execution workflow automatically pauses at critical steps. Action proceeds only upon explicit human review and approval.



Human-on-the-Loop (HOTL)

The AI executes actions autonomously. The human operator monitors the stream and retains real-time override/veto authority.



Human-out-of-the-Loop

The system is completely self-governing within set operational boundaries. Restricted to non-critical and low-risk domains.

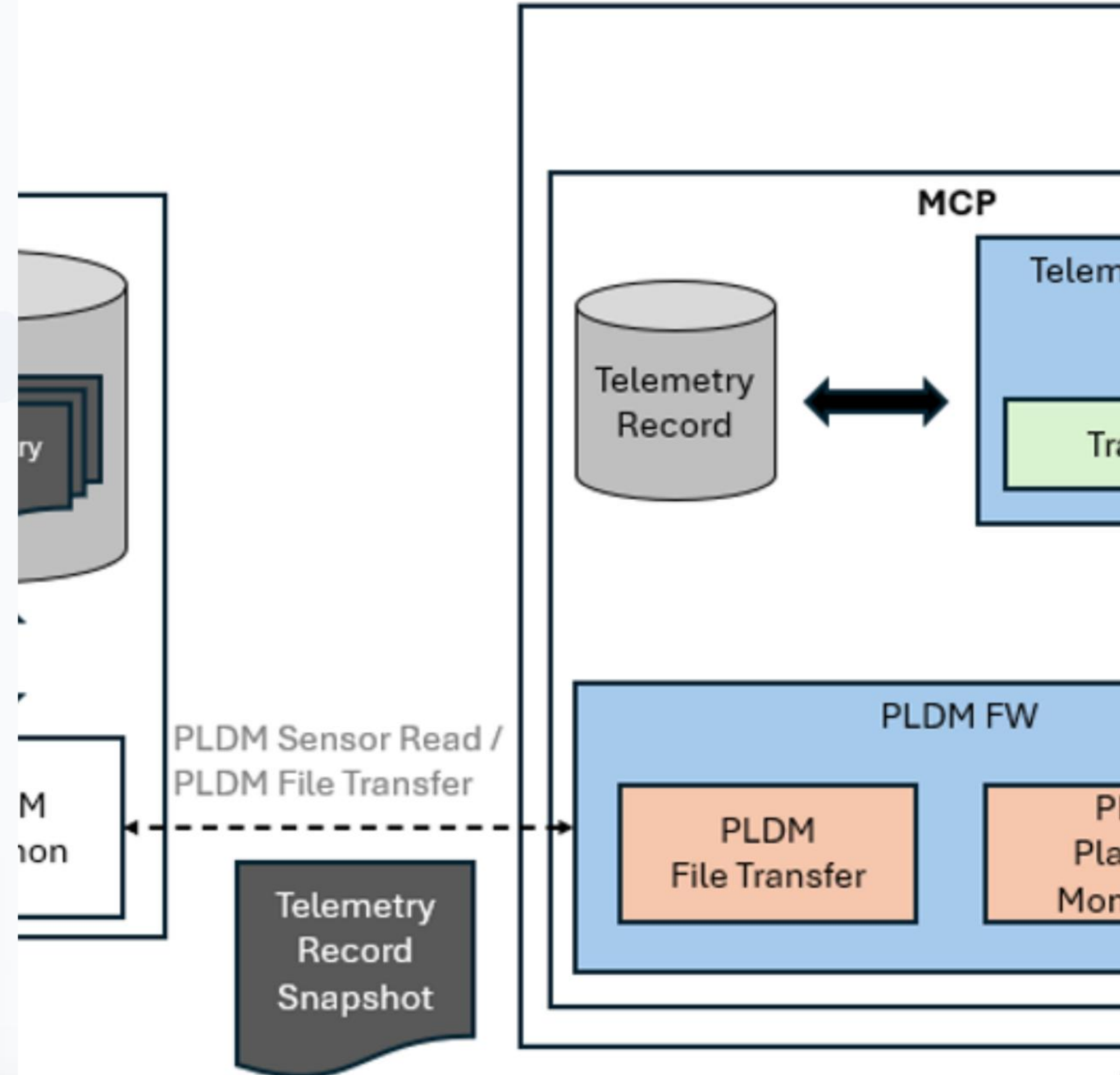
The Infrastructure of Oversight

State Persistence & SLAs

To safely pause agent workflows for human review, systems must execute state serialization checkpoints. This saves the memory, history, and tool outputs before pausing execution.

$$\text{Latency}_{\text{Total}} = \text{Latency}_{\text{Model}} + \text{Latency}_{\text{Oversight}}$$

Decouple synchronous requests from async human pipelines to prevent severe latency degradation across agentic tool chains.



SECTION 05

Multi-Dimensional Evaluation

A multidimensional evaluation blueprint testing across qualitative, quantitative, intrinsic, and longitudinal metrics.

Multi-Dimensional Evaluation Matrix

How & What (Execution Logs)

- ✓ **Quantitative vs. Qualitative:** Measuring transaction rates & error counts vs. understanding subjective friction points.
- ✓ **Intrinsic vs. Extrinsic:** Evaluating model outputs in isolated sandboxes vs. tracking real-world collaborative success.
- ✓ **Edit Traces:** Analyzing granular structural logs of user edits on system text outputs to map capability gaps.

Who & When (Socio-Temporal)

- ✓ **Target Demographics:** Custom testing strategies targeting Domain Experts, Lay Users, and LLM-Simulated Persona clusters.
- ✓ **Short vs. Longitudinal:** Isolating initial "Novelty Bias" spikes from durable, multi-week software adoption rates.

⚠ **Framing Bias:** Positive/negative survey structure heavily skews metric outcomes.

SECTION 06

Ethics, Compliance, & Trustworthy AI

Integrating IBM's Ethical Pillars, complying with the EU AI Act mandates, and deploying open-source testing libraries.

| IBM's Five Pillars of Trustworthy AI

- ✓ **Explainability:** Inner computation paths and feature weights must be fully transparent and understandable to human system monitors.
- ✓ **Fairness:** Teams must run systematic biases audits across all datasets and models to detect and mitigate unwanted discriminatory skew.
- ✓ **Robustness:** System boundaries must protect against malicious exploits, evasion hacks, and prompt-injection vectors.
- ✓ **Transparency:** Provide comprehensive public-facing disclosures describing pipeline details, training histories, and data parameters.
- ✓ **Privacy:** Enforce strict, local privacy boundaries while granting users total operational control over their personal telemetry data.

| Compliance & Actionable Developer Toolkits

THE EU AI ACT (ARTICLE 14)

High-risk models must legally enforce built-in **human oversight mechanisms**. Systems must afford operators full visibility into constraints, anomaly alerts, and a physical/logical emergency override (kill switch).

IBM OPEN-SOURCE ECOSYSTEM

- ✓ **AI Fairness 360 (AIF360):** Checks and mitigates algorithmic bias in ML.
- ✓ **Adversarial Robustness Toolbox (ART):** Defends networks against data extraction, poisoning, and evasion.
- ✓ **AI FactSheets 360:** Standardized "nutrition label" documentation for lifecycle tracking.

 Futuristic secure digital shield representing trustworthy artificial intelligence and ethical standards

THANK YOU FOR YOUR TIME

Questions & Discussion

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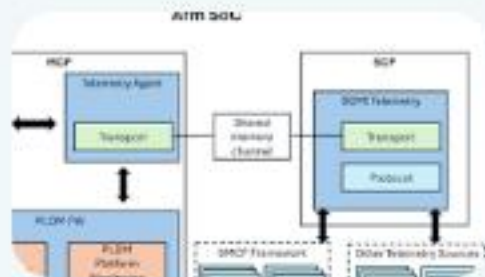
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